

7.3 Reformulation of part II of the proof of Lemma 3.3

Lemma 7.6 (From small density to small density ratio). *Suppose $E \subset \mathbb{R}^n$, $x \in E$ and*

$$\limsup_{r \rightarrow 0} \frac{\mathcal{H}^s(B_r(x) \cap E)}{(2r)^s} < \frac{1}{2^s}.$$

Then there exists $\delta > 0$ so that

$$\frac{\mathcal{H}^s(B_r(x) \cap E)}{(2r)^s} < \frac{1 - \delta}{2^s}$$

for every $r \in (0, \delta]$.

Proof. By Lemma 4.2 applied with $f(r) = \frac{\mathcal{H}^s(B_r(x) \cap E)}{(2r)^s}$ and $M = \frac{1}{2^s}$, there exist $\varepsilon > 0$ and $R > 0$ such that for all $r \in (0, R]$,

$$\frac{\mathcal{H}^s(B_r(x) \cap E)}{(2r)^s} < \frac{1}{2^s} - \varepsilon. \quad (7.10)$$

Let $\delta = \min\{R, 2^s \varepsilon\}$. Then $\frac{\delta}{2^s} \leq \varepsilon$, and for all $r \in (0, \delta]$ we have $r \in (0, R]$, so by (7.10),

$$\frac{\mathcal{H}^s(B_r(x) \cap E)}{(2r)^s} < \frac{1}{2^s} - \varepsilon \leq \frac{1 - \delta}{2^s}.$$

□

Lemma 7.7. *If $\delta > 0$, $C, E \subset \mathbb{R}^n$, $x \in E \cap C$, $\text{diam}(C) \leq \delta$ and*

$$\frac{\mathcal{H}^s(B_r(x) \cap E)}{(2r)^s} < \frac{1 - \delta}{2^s},$$

for every $r \in (0, \delta]$, then

$$\mathcal{H}^s(C \cap E) \leq (1 - \delta)|C|^s.$$

Proof. Since $x \in C$, we have $C \subset B_{|C|}(x)$, so

$$\mathcal{H}^s(C \cap E) \leq \mathcal{H}^s(B_{\text{diam}(C)}(x) \cap E). \quad (7.11)$$

By assumption, taking $r = \text{diam } C$ and using that $\text{diam}(C) \leq \delta$, we get

$$\mathcal{H}^s(B_{\text{diam}(C)}(x) \cap E) \leq (1 - \delta)\text{diam}(C)^s. \quad (7.12)$$

Combining (7.11) and (7.12) we obtain

$$\mathcal{H}^s(C \cap E) \leq (1 - \delta)\text{diam}(C)^s.$$

□

Lemma 7.8. *Suppose $\mathcal{H}^s(E) < \infty$. Then*

$$\left\{ x \in E : \limsup_{r \rightarrow 0} \frac{\mathcal{H}^s(B_r(x) \cap E)}{(2r)^s} < \frac{1}{2^s} \right\} \subset \bigcup_{k=1}^{\infty} E \left(\frac{1}{k}, 1 - \frac{1}{k} \right). \quad (7.13)$$

Proof. Let $x \in \{x \in E : \limsup_{r \rightarrow 0} \frac{\mathcal{H}^s(B_r(x) \cap E)}{(2r)^s} < \frac{1}{2^s}\}$. By Lemma 7.6, there exists $\delta > 0$ such that for all $r \in (0, \delta]$,

$$\frac{\mathcal{H}^s(B_r(x) \cap E)}{(2r)^s} < \frac{1 - \delta}{2^s}.$$

Let $C \subset \mathbb{R}^n$ be such that $x \in C$ and $\text{diam}(C) \leq \delta$. By Lemma 7.7, we have

$$\mathcal{H}^s(B_{\text{diam } C}(x) \cap E) \leq (1 - \delta)(\text{diam } C)^s,$$

so $x \in E(\text{diam } C, 1 - \text{diam } C)$. Let $k \in \mathbb{N}$ be such that $\frac{1}{k} \leq \text{diam } C$. Then by 7.1 we have $E(\text{diam } C, 1 - \text{diam } C) \subset E(\frac{1}{k}, 1 - \frac{1}{k})$, so $x \in E(\frac{1}{k}, 1 - \frac{1}{k})$. $\square$

Lemma 7.9 (Part 2 of Lemma 3.3). *Suppose $\mathcal{H}^s(E) < \infty$. Then*

$$\mathcal{H}^s \left(\left\{ x \in E : \limsup_{r \rightarrow 0} \frac{\mathcal{H}^s(B_r(x) \cap E)}{(2r)^s} < \frac{1}{2^s} \right\} \right) = 0. \quad (7.14)$$

Proof. By Lemma 7.8, sub-additivity of $\mathcal{H}^s$ and Lemma 7.5, we have

$$\mathcal{H}^s \left(\left\{ x \in E : \limsup_{r \rightarrow 0} \frac{\mathcal{H}^s(B_r(x) \cap E)}{(2r)^s} < \frac{1}{2^s} \right\} \right) \leq \sum_{k=1}^{\infty} E \left(\frac{1}{k}, 1 - \frac{1}{k} \right) = 0,$$

which implies (7.14). $\square$